

Assignment 1: Dataset Proposal

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Dataset Name and Source

Dataset name: Brazilian E-Commerce Public Dataset by Olist

Source URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Published by Olist, Brazil's largest department-store marketplace aggregator, for public academic and research use under Kaggle's dataset terms. The data is real, anonymised commercial transaction data collected between October 2016 and September 2018.

Row Count and File Size

The dataset is a relational collection of nine CSV files linked by shared identifier columns. Combined, the core tables total over 400,000 rows prior to any join, filter, or aggregation.

File	Approx. Rows	Role
olist_orders_dataset.csv	~99,441	Core order-level fact table
olist_order_items_dataset.csv	~112,650	Line-item level (order × product × seller)
olist_order_payments_dataset.csv	~103,886	Payment transactions per order
olist_order_reviews_dataset.csv	~99,224	Customer review per order
olist_customers_dataset.csv	~99,441	Customer dimension
olist_products_dataset.csv	~32,951	Product dimension
olist_sellers_dataset.csv	~3,095	Seller dimension
olist_geolocation_dataset.csv	~1,000,000	Zip-code-prefix
product_category_name_translation.csv	71	Category name in English translation

Total compressed size: approximately 45–50 MB (zip download); approximately 90 MB uncompressed CSV.

Column Inventory

Six of the nine tables will be used directly in the analytical pipeline. Columns are listed with their data type and a brief description.

olist_orders_dataset.csv

Column	Type	Description
order_id	string	Unique order identifier (primary key)
customer_id	string	Foreign key to the customers table
order_status	string	delivered / shipped / canceled / etc.
order_purchase_timestamp	timestamp	When the order was placed
order_approved_at	timestamp	When payment was approved
order_delivered_carrier_date	timestamp	Handover to logistics partner
order_delivered_customer_date	timestamp	Actual delivery date

order_estimated_delivery_date	<i>timestamp</i>	Promised delivery date
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olist_order_items_dataset.csv

Column	Type	Description
order_id	<i>string</i>	Foreign key to orders
order_item_id	<i>integer</i>	Sequence number within the order
product_id	<i>string</i>	Foreign key to products
seller_id	<i>string</i>	Foreign key to sellers
shipping_limit_date	<i>timestamp</i>	Seller's shipping deadline
price	<i>double</i>	Item price
freight_value	<i>double</i>	Shipping cost

olist_order_payments_dataset.csv

Column	Type	Description
order_id	<i>string</i>	Foreign key to orders
payment_sequential	<i>integer</i>	Payment attempt sequence
payment_type	<i>string</i>	credit_card / boleto / voucher / debit_card
payment_installments	<i>integer</i>	Number of instalments
payment_value	<i>double</i>	Amount paid

olist_customers_dataset.csv

Column	Type	Description
customer_id	<i>string</i>	Order-scoped customer key (primary key)
customer_unique_id	<i>string</i>	True unique customer identifier
customer_zip_code_prefix	<i>string</i>	First five digits of postal code
customer_city	<i>string</i>	Customer city
customer_state	<i>string</i>	Brazilian state (categorical, 27 values)

olist_products_dataset.csv

Column	Type	Description
product_id	<i>string</i>	Unique product identifier (primary key)
product_category_name	<i>string</i>	Category name (Portuguese)
product_weight_g	<i>double</i>	Product weight
product_length_cm / height_cm / width_cm	<i>double</i>	Product dimensions

product_category_name_translation.csv

Column	Type	Description
product_category_name	<i>string</i>	Portuguese category name (join key)
product_category_name_english	<i>string</i>	English translation

High-Cardinality Column for Partitioning Experiments

Partitioning experiments require a column with enough distinct values to distribute meaningfully across multiple partitions. From `olist_order_items_dataset.csv`, two candidates stand out: `product_id`, ~33,000 distinct values across ~112,650 rows and `order_id` ~99,441 near-unique values, since most orders contain a single line item.

`product_id` will be used as the primary partitioning key. Its distribution is naturally skewed as a small number of bestselling products account for a disproportionate share of order-item rows, while most products sell only a handful of times. This gives the hash-partitioning versus range-partitioning comparison a genuine skew pattern to reveal, rather than a synthetic or evenly distributed one. `order_id`, being closer to uniformly distributed, is retained as a secondary comparison point to contrast a skewed key against a near-uniform one.

Proposed Business Case

This analysis will examine revenue contribution and delivery performance across product categories, Brazilian states, and time. Specifically, it will identify which product categories generate the highest revenue within each customer state per quarter ranked using a window function, filtered to categories exceeding a minimum order-volume threshold. It will also investigate whether late deliveries, measured as the gap between estimated and actual delivery dates correlate with lower customer review scores, and whether this relationship differs by category or region. This directly informs commercial questions such as which category–region combinations are underperforming due to logistics issues versus genuine demand, and where operational investment would have the greatest impact.

Distributed Analytics

This dataset, and the proposed analysis, is well suited to distributed processing and the scope of simple filtering or a single GROUP BY operation.

The analysis requires multi-way joins, not a single join. Answering the business question requires joining `order_items` ⋈ `orders` ⋈ `customers` ⋈ `products` ⋈ `category_translation` which is a minimum four-way join across tables of substantially different sizes, ranging from 71 rows (`category_translation`) to approximately 112,650 rows (`order_items`). This size disparity is the scenario in which join has a material effect on performance, allowing for further investigation.

The core query depends on a window function, not a plain aggregation. Ranking categories by revenue within each state–quarter partition (`Window.partitionBy("state", "quarter").orderBy(revenue.desc())`) to surface the top-performing categories per group is structurally different from, and more complex than, a single GROUP BY.

The query combines aggregation with a derived time dimension and a post-aggregation filter. State–category combinations with fewer than a minimum number of orders are excluded after aggregation — a HAVING-equivalent step that a single-stage GROUP BY cannot express on its own.

The delivery-delay sub-question requires a derived column computed prior to aggregation. Delivery delay (actual delivery date minus estimated delivery date) must be calculated at the row level before it can be aggregated or correlated with review scores, adding a genuine transformation step ahead of the grouping stage.

The dataset exhibits natural data skew. Skew in `product_id` and `seller_id` — a small number of high-volume products and sellers alongside a long tail of low-volume ones — provides an authentic, rather than artificially constructed, basis for the partitioning and skew-analysis work required in the assignment.

Assessment 2 ML Analysis

An initial consideration for Assessment 2, several columns in the dataset naturally support later machine learning work:

- **Numeric features:** `price`, `freight_value`, `payment_value`, `payment_installments`, `product_weight_g`, product dimensions, and delivery delay (a derived feature from the order date fields).
- **Categorical features:** `product_category_name`, `customer_state`, `payment_type`, `order_status`.
- **Target candidates:** `review_score` (from the reviews table, if incorporated), supporting a classification or regression task predicting customer satisfaction; or delivery delay itself as a regression target.

These features support plausible future directions such as predicting customer review score or delivery delay from order, product, and payment attributes.